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COLLEGE OF ENGINEERING AND TECHNOLOGY
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
KATTANKULATHUR- 603203

Title of Exercise : **PUBLICATION ETHICS AND MISCONDUCT**

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Register Number : RA2512019010001

Date of Exercise :

S.No	Marks Split Up	Maximum Marks	Marks Obtained
1	Concept Identification	4	
2	Explanation	4	
3	Application of Knowledge/ Awareness of Ethical Guidelines	4	
4	Analytical / Quantitative Skill	4	
5	Critical Thinking / Evaluation	4	
	Total	20	

Staff Signature

Question-1:

A researcher modifies a dataset to fit a hypothesis that predicts a 20% increase in battery efficiency.

- a) Identify which type of misconduct this represents (Falsification, Fabrication, or Plagiarism).
- b) If the original dataset had 200 measurements and 50 were altered, calculate the percentage of falsified data.

Solution:

a) This action represents Falsification. It occurs when a researcher changes or manipulates actual research data so that the findings match an expected outcome, instead of reporting the measurements as they were originally observed.

Falsification is the act of manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record. In this case, modifying the dataset to fit a predetermined hypothesis (a 20% increase in battery efficiency) is a direct manipulation of the data.

b)

Total measurements = 200

Modified measurements = 50

$$\text{Falsified percentage} = \frac{50}{200} \times 100 = 25\%$$

Hence, 25% of the dataset was falsified.

Question-2:

A researcher submits the same results to two journals to increase publication count.

- a) Define duplicate publication and salami slicing.
- b) Explain why these practices violate publication ethics.

Solution:

a) Definitions:

Duplicate Publication:

- Submitting the same research results to more than one journal without proper disclosure. It misrepresents the uniqueness of the study and artificially increases publication numbers.
- Duplicate publication is the act of re-publishing the same research, results, or analysis in more than one journal without informing the editors. It presents previously published work as new, which is misleading and violates the requirement for originality in academic publishing.

Salami Slicing:

- Dividing one complete research work into several smaller papers just to increase the number of publications, even when the pieces do not stand independently.
- Instead of presenting the research as one coherent article, the author spreads fragmented results across multiple papers to increase publication numbers.

b) These practices break publication ethics because they mislead the scientific community, create unnecessary duplicate work for reviewers and editors, and distort the true contribution of a researcher. Duplicate publication wastes editorial resources, while salami slicing weakens the scientific value of research by splitting meaningful findings into fragments.

Question–3

A journal editor receives a complaint of suspected plagiarism in a submitted manuscript.

a) Name two international guidelines or organisations that help maintain publication ethics (e.g., COPE, WAME).

b) Explain the steps the editor should follow according to these guidelines.

Solution:

a) Two international bodies that promote ethical publishing are:

1. **COPE** – Committee on Publication Ethics
2. **WAME** – World Association of Medical Editors

b) According to the guidelines of organisations like COPE and WAME, the editor should:

1. Acknowledge the complaint and examine the allegation to check if plagiarism might be present.
2. Contact the author to request clarification about the overlapping content.
3. Compare the manuscript with the original source to determine the extent of copied material.
4. Decide the next step:
 - If the issue is discovered before publication, the editor may reject the manuscript and inform the author's institution.
 - If it is found after publication, a correction or retraction must be issued.
5. Maintain proper documentation of the investigation and ensure that fairness, confidentiality, and transparency are followed throughout the process.

Question–4

A researcher receives an email offering quick publication in a “high-impact journal” for a fee of \$1,500 and no peer review.

- a) Identify why this is considered a predatory journal.**
- b) Explain two strategies to verify the credibility of a journal before submission.**

Solution:

a) This is classified as a predatory journal because it demands money, promises unrealistic publication speed, and ignores the standard peer-review process. Predatory journals focus on profit rather than scientific quality or ethical publishing.

b) Strategies to confirm a journal’s authenticity:

1. Check indexing and recognition:
Look for the journal in standard databases such as Scopus, Web of Science, or DOAJ. Also verify the publisher’s membership in ethical bodies like COPE.
2. Inspect the journal’s website:
Review the clarity of the peer-review process, editorial board details, publication policies, and contact information. Legitimate journals provide transparent and verifiable details, whereas predatory journals often offer vague or misleading information.

Question–5

A student’s thesis report shows a similarity index of 15% when checked with plagiarism-detection software.

- a) Explain whether this similarity is acceptable and why.**
- b) Suggest steps to reduce plagiarism if it exceeds acceptable limits.**

Solution:

a) A similarity level of 15% is generally acceptable in many academic institutions. Some standard phrases and referenced content naturally appear in similarity reports, as long as they are properly cited and do not involve copying unique ideas or original analysis.

b) If the similarity exceeds recommended limits, the student can reduce it by:

1. **Paraphrasing correctly:** Rewrite ideas in their own words while preserving the original meaning.
2. **Adding citations:** Give proper credit to all sources of information, data, and quotations.
3. **Using plagiarism-checking tools:** Review drafts and correct any unintentional overlap before submission.

4. **Avoiding self-plagiarism:** Do not reuse large portions of previous work without rephrasing and citing appropriately.

Inference

Ethical publishing requires honesty, originality, and proper citation. Researchers must avoid falsifying data, duplicating work, or submitting to predatory journals. Following guidelines like COPE and WAME helps ensure integrity and supports reliable scientific progress.